William Trottier

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Work Experience

Software Developer / AI Trainer (Contract) — December 2024 – January 2026

DataAnnotation – Remote

- Developed RESTful APIs, leveraged existing APIs, and designed JSON schemas for the API calls to facilitate the training of large-language models for optimal performance within specific application contexts
- Created coding challenges for 5+ projects for chatbot development encompassing various programming concepts and evaluated chatbot response for accuracy, efficiency, and adherence to best practices
- Trained LLMs to interface with and control mobile applications while encouraging parallelization for efficiency

Teachers Assistant — May 2024 – August 2024

University of Victoria – Victoria, BC

- Prepared and delivered labs to 2 30-student lab groups weekly, alternating between theory and programming in which an initial presentation was delivered and a final solution walkthrough
- Collaborated in a small team to manage a 2nd year 100-student algorithms course, including marking assignments and examinations, and providing timely feedback

Education

University of Victoria — December 2024

Bachelor of Science, Computer Science: Graphics and Gaming Specialization, **GPA: 3.9**

Projects

Cumulus Cloud Animation by Continuous Cellular Automata - *github.com/williamt1117/CloudAnimation*

- Designed an animation system in Unity using C# that operates on a 3D density field representing clouds that utilizes cellular automata with a focus on unique cloud formation and dissipation to replicate nimbus clouds
- Conducted research on various methods of cloud simulation and produced an accompanying academic report

Ransomware Data Visualization and Analytics - *github.com/williamt1117/RansomwareDataVisualization*

- Utilized D3 JavaScript library to create an explorable circle packing diagram with up to 3 layers of grouping with supporting histogram and pie chart to analyze premeditated questions and trends about the data
- Collaborated with a small team to split work and present a demonstration of the project and findings

METAR Map

- Developed a C++ Arduino program for a ESP8266 based microcontroller to fetch real-time weather and twilight data via HTTP requests and APIs via a Wi-Fi enabled processor and control an addressable LED chain
- Performed soldering of LED and board pin connections to ensure proper functionality

Ray Tracer - *github.com/williamt1117/CppRayTracer*

- Constructed a backwards ray tracer in C++ that renders a scene and outputs images from processed text files, incorporating the Phong illumination model to illuminate a complex multi-sphere environment
- Implemented reflectivity by recursive ray tracing that supports depths of 10+ bounces

Tetris AI - *github.com/williamt1117/Tetris-AI*

- Developed an AI system in Unity using C#, which generates sub-states of the game board to evaluate move-sets
- Employed a genetic-learning algorithm to optimize the AI, which performed 100+ iterations to enhance its ability to efficiently clear lines in time and space-constrained scenarios

Skills

Languages: C, C#, Java, PostgreSQL, Python
Software: Arduino IDE, Git, SolidWorks, Unity